VIET NAM NATIONAL UNIVERSITY HO CHI MINH CITY
UNIVERSITY OF ECONOMICS AND LAW

FINAL PROJECT PROPOSAL

AI-BASED SOLUTION IN BUSINESS AND MANAGEMENT

AN EXPLAINABLE AI SYSTEM FOR VERIFYING PLASTIC PACKAGING RECYCLING CLAIMS IN FMCG SUSTAINABILITY REPORTS

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INTRODUCTION

In the context of growing environmental awareness and increasing corporate commitments to sustainability, Fast-Moving Consumer Goods (FMCG) companies are under pressure to reduce packaging waste and improve the recyclability of their products. Sustainability reports have become an important communication channel through which companies disclose environmental performance, recycling initiatives, and packaging-related commitments. However, environmental claims may not always be accompanied by sufficient evidence, clearly defined measurement boundaries, or consistent quantitative information. This creates challenges for stakeholders seeking to assess the credibility of corporate sustainability disclosures.

This project proposes an Explainable Artificial Intelligence (XAI) system to verify plastic packaging recycling claims in publicly available sustainability reports published by FMCG companies. The system will use Natural Language Processing (NLP) and Large Language Models (LLMs) to identify relevant claims, retrieve supporting evidence from source documents, and assess whether the evidence adequately supports each claim. In addition, rule-based quantitative validation will be applied when sufficient numerical data are available to identify potential inconsistencies in reported recycling figures and rates.

The project aims to demonstrate how AI can support more transparent and evidence-based sustainability assessment. Rather than automatically determining whether a company engages in greenwashing, the proposed system will identify claims that require further verification, explain the reasons for each assessment, and provide traceable references to the original documents.

DESCRIPTION OF THE PROJECT

This project focuses on developing an AI-assisted system for verifying claims related to plastic packaging recycling in FMCG sustainability reports. The system will analyze both textual statements and relevant numerical indicators to identify insufficient supporting evidence, mismatches between claims and evidence, and potential inconsistencies in reported figures.

The proposed system consists of five main stages:

1. **Data Collection and Preprocessing**

The project will collect publicly available sustainability reports, ESG reports, and related corporate disclosures from selected FMCG companies. Documents will be processed to extract textual content, tables, page numbers, and relevant metadata. The scope will be restricted to claims concerning plastic packaging recyclability, recycling performance, and related commitments.

2. **Claim Extraction and Classification**

NLP and LLM techniques will be used to identify and extract relevant environmental claims. Each claim will be represented using structured attributes, including the claim text, reporting year, packaging type, numerical value (if applicable), scope, and source location. This step will transform unstructured sustainability disclosures into a structured dataset for subsequent verification.

3. **Evidence Retrieval and Semantic Verification**

NLP and LLM approach will retrieve relevant passages, tables, and supporting information from the collected documents. The system will compare each claim with its retrieved evidence to assess semantic relevance, scope alignment, and the availability of supporting information. Claims will be assigned preliminary assessment labels, such as:

- **Supported:** The retrieved evidence adequately supports the claim within the available context.

- **Insufficient Evidence:** The available documents do not provide enough information to verify the claim.
- **IScope Mismatch:** The evidence covers a narrower or different scope than the claim.
- **Potential Inconsistency:** The claim conflicts with other information identified in the available documents.

These labels are intended to support human review and do not constitute definitive judgments of corporate greenwashing.

4. Rule-Based Quantitative Validation

When sufficient numerical information is available, the system will apply rule-based checks to validate reported recycling figures. These checks may include unit normalization, percentage recalculation, consistency of denominators, and comparisons between collected and recycled packaging quantities.

For example, if a company reports 10,000 tonnes of packaging placed on the market, 4,000 tonnes collected, and 2,500 tonnes recycled, the system can calculate the corresponding collection and recycling rates, provided that the figures share the same reporting scope and denominator. It can then flag discrepancies between the reported and recalculated values for further examination.

This component will be optional at the claim level because publicly available reports may not disclose all the information required for quantitative verification.

5. Explainable Results and Prototype Development

The system will present verification results through an interactive web interface. For each claim, users will be able to review the assessment label, the reason for the assessment, relevant supporting passages, source page numbers, and any detected numerical inconsistencies.

The prototype will prioritize traceability and interpretability, allowing users to examine the evidence behind each result rather than relying solely on an AI-generated conclusion.

POTENTIAL DATA SOURCES

The project will prioritize public and accessible data sources to ensure feasibility within the project timeline.

Data Source	Description	Language	Usage Purpose
Corporate Sustainability Reports	Public sustainability or ESG reports from selected FMCG companies containing packaging, waste, and recycling disclosures	English and/or Vietnamese	Primary source for claim extraction and evidence retrieval
Corporate Environmental Disclosures	Public documents providing supplementary information about packaging materials, recycling initiatives, and sustainability commitments	English and/or Vietnamese	Supplementary evidence for claim verification
GRI Standards	Relevant sustainability reporting guidance, particularly GRI 306 (Waste) and, where applicable, GRI 301 (Materials)	Primarily English	Inform the design of extraction fields and verification criteria

Potential companies may include Vinamilk, Masan Consumer, and Unilever, subject to the availability and relevance of their public reports. The final selection will depend on whether the documents contain sufficient information about plastic packaging and recycling.

TENTATIVE TECHNIQUES AND MODELS TO BE APPLIED

Component	Potential Technique	Purpose
Document Processing	PDF text extraction; OCR when necessary	Extract text and relevant information from sustainability reports
Text Preprocessing	Text cleaning, segmentation, metadata extraction	Prepare document content for subsequent analysis
Claim Extraction	LLM-based structured extraction; NLP techniques	Identify plastic packaging recycling claims and convert them into structured records
Evidence Retrieval & Verification	LLM-based comparison with predefined criteria	Assess evidence sufficiency, semantic alignment, and scope mismatch
Quantitative Validation	Python, Pandas, and rule-based calculations	Recalculate rates and identify numerical or unit inconsistencies
Explainability	Source-grounded explanations and page-level references	Provide traceable reasons and supporting evidence for each assessment
Prototype Development	Streamlit	Build an interactive interface for uploading documents and reviewing results
Evaluation	Precision, Recall, F1-score, and retrieval evaluation	Measure claim extraction, classification, and evidence retrieval performance

DELIVERABLES OF THE PROJECT

Deliverable	Description
1. Processed Dataset	A structured dataset containing extracted recycling claims, relevant evidence, source references, and human-assigned labels
2. Python Notebooks / Scripts	Implementation of document processing, claim extraction, evidence retrieval, verification, and quantitative validation
3. AI-Based Web Prototype	A Streamlit application that displays claim-level assessments, explanations, supporting passages, and source page references
4. Analytical Report	A report describing the problem, data sources, methodology, evaluation results, limitations, and business implications
5. Presentation Slides	A concise presentation of the proposed solution, technical pipeline, findings, and prototype demonstration

EXPECTED OUTCOMES AND BUSINESS VALUE

The project is expected to produce a functional prototype that helps users examine plastic packaging recycling claims in FMCG sustainability reports more systematically. By combining semantic evidence retrieval with rule-based numerical checks, the system will demonstrate how AI can improve the traceability and efficiency of preliminary sustainability claim verification.

Potential users include sustainability analysts, ESG researchers, corporate reporting teams, and stakeholders who review environmental disclosures. The prototype may help them identify claims requiring additional documentation, reduce the effort needed to locate relevant evidence, and detect discrepancies that warrant further investigation.

The project is subject to several limitations. Publicly available reports may provide incomplete information, terminology and measurement boundaries may vary across companies, and automated language models may produce incorrect interpretations. Therefore, the system will provide evidence-linked assessments for human review rather than definitive accusations of greenwashing.

Overall, this project aims to demonstrate a feasible and explainable AI solution for supporting the credibility assessment of sustainability disclosures, with a clearly defined focus on plastic packaging recycling in the FMCG industry.